
SYSTEMATIC REVIEW PROTOCOL / 1-MONTH PLAN

Factuality in Clinical Text Simplification

*A 1-Month Systematic Review Protocol for Low-Resource
Bangla Healthcare NLP*

PRISMA 2020 + PICO/SPIDER + PROSPERO + Cochrane RoB 2/ROBINS-I
+ GRADE + PRISMA-S. Target venue: Journal of Biomedical Informatics
(Q1, Elsevier).

Research Plan

Bangla Clinical NLP Initiative / 2026

PRISMA 2020

PICO + SPIDER

GRADE

Table of Contents

1. Background and Rationale

2. Review Scope and Objectives

3. Research Questions (PICO + SPIDER)

4. PROSPERO Pre-Registration Plan

5. Search Strategy (PRISMA-S)

6. Inclusion and Exclusion Criteria

7. Screening Workflow and PRISMA 2020 Flow

8. Quality and Risk-of-Bias Assessment

9. Data Extraction Codebook

10. Synthesis Methods (SWiM + GRADE)

11. Gap Taxonomy

12. Inventory of Datasets (Realistically Available)

3

3

4

5

6

7

8

8

9

10

10

11

13. Inventory of Methods and Baselines	12
14. Expected Contributions of the Review	13
15. Four-Week Execution Timeline	13
16. Risk Register and Mitigations	13
17. Target Venue and Submission Package	14
Submission Checklist	14
18. Expected Review Paper Outline	15
19. References (Anchor Papers)	15

1. Background and Rationale

Clinical documents such as discharge summaries, pathology reports, prescriptions, and laboratory results are written in highly technical language that most patients cannot understand. In Bangladesh, the gap is sharper still: documentation is predominantly in English while the patient population speaks Bangla, and often reads at a primary-school level. The resulting communication gap directly affects patient awareness, treatment compliance, and downstream health outcomes. Recent advances in transformer-based natural language processing (NLP) have enabled significant progress in text simplification for high-resource languages, yet most of this work has been evaluated on English clinical text with little attention to factual faithfulness or to the cross-lingual transfer to low-resource languages such as Bangla.

A second, equally important problem is that simplification models are prone to hallucinating or rewriting critical medical facts. The risk is acute when models are trained on synthetic data generated by other large language models, creating a closed synthetic loop where the system learns artifacts of the generator rather than genuine clinical language. Prior work has also conflated entity preservation with factuality: a simplification that retains the tokens "Metformin" and "kidney failure" but inverts their relation (e.g., "continue" instead of "stop") has perfect lexical overlap yet catastrophic factual failure. This proposal therefore treats factuality preservation, low-resource adaptability, and cultural appropriateness as a joint research problem rather than three independent sub-tasks.

The review paper planned in this document is the first deliverable of a two-paper research arc. The second deliverable, planned in the companion journal plan, will empirically test the gaps identified here through a hybrid Clinical Fact Graph Preservation (CFGP) + Medical NLI Gate + Cross-Lingual KG-RAG framework. This review is therefore scoped tightly to map the factuality-preservation landscape, inventory the available Bangla resources, and derive a defensible set of research questions and gaps that the empirical study will address.

Working definition. "Factuality" in this review denotes the preservation of clinical meaning across three levels: (i) entities (drugs, diseases, procedures), (ii) relations between entities (treats, causes, contraindicates), and (iii) attributes (dosage, frequency, temporal, negation). Surface-level lexical overlap is explicitly treated as insufficient evidence of factuality.

2. Review Scope and Objectives

The review is bounded by three concentric circles that are kept explicit so the heterogeneous literature does not collapse into an unmanageable set. The core circle covers transformer-based medical and clinical text simplification in any language, with an English-dominant literature expected. The first adjacent circle widens to multilingual, cross-lingual, and low-resource generation methods applied to clinical or biomedical text, including clinical machine translation and biomedical natural language generation in low-resource settings, even when these are not strictly simplification tasks. The second adjacent circle inventories Bangla NLP for healthcare specifically, covering any task (named entity recognition, summarization, translation, optical character recognition) so that the existing infrastructure can be mapped.

Primary Objective

To systematically map the current state of transformer-based approaches for generating patient-friendly explanations from clinical and biomedical text, and to assess the extent to which these approaches are adaptable to low-resource, multilingual settings such as Bangla, with explicit attention to factuality preservation mechanisms.

Secondary Objectives

- **O1.** Build a taxonomy of factuality-preservation mechanisms used in medical text simplification, ranging from lexical constraints through relation-level graph preservation to natural language inference gates.
- **O2.** Inventory all publicly available datasets relevant to clinical text simplification and Bangla healthcare NLP, with size, license, construction method, and suitability for factuality research.
- **O3.** Catalogue the evaluation methods used to measure factuality, readability, and patient comprehension, identifying under-powered studies and missing baselines.
- **O4.** Identify and clearly articulate the gaps that a follow-up empirical study must address, providing a defensible bridge from the review to the companion journal paper.

3. Research Questions (PICO + SPIDER)

The review adopts a dual-framework strategy. PICO (Population, Intervention, Comparison, Outcome) structures the quantitative research questions about model architectures and evaluation outcomes. SPIDER (Sample, Phenomenon of Interest, Design, Evaluation, Research type) structures the qualitative questions about dataset construction, annotation practice, and cultural appropriateness. This dual approach is recommended for mixed-methods systematic reviews in health informatics.

PICO Frame (Quantitative RQs)

Element	Definition
Population	Clinical, biomedical, or pathology text intended for patient comprehension: discharge summaries, pathology/radiology reports, medication instructions, lay-summarized literature.
Intervention	Transformer-based methods (BERT, T5, BART, GPT-family, LLMs) for simplification, summarization, explanation generation, including fine-tuning, prompting, retrieval-augmented generation, and constrained decoding.
Comparison	Non-transformer baselines (rule-based, statistical), multilingual vs. monolingual, human-expert gold standard, factuality-constrained vs. unconstrained decoding.
Outcome	Readability scores (FKGL, SMOG, Bangla-adapted), factual and medical accuracy preservation, human evaluation scores, low-resource transferability metrics.
Context	Academic and clinical NLP venues, 2018-2026, any source language with explicit Bangla or South Asian sub-analysis where available.

Table 3.1 - PICO frame for the quantitative research questions.

SPIDER Frame (Qualitative RQs)

Element	Definition
Sample	Peer-reviewed and vetted preprint studies of clinical text simplification or Bangla healthcare NLP, 2018-2026.
Phenomenon of Interest	Dataset construction practice (expert-annotated vs. synthetic), factuality evaluation methodology, cultural adaptation strategies.
Design	Systematic review with narrative synthesis under SWiM guidelines; meta-analysis not feasible due to heterogeneity.
Evaluation	Comparative appraisal of factuality preservation mechanisms, dataset characteristics, and evaluation rigor.
Research type	Mixed-methods systematic review (quantitative mapping + qualitative thematic synthesis).

Table 3.2 - SPIDER frame for the qualitative research questions.

Research Sub-Questions

ID	Sub-Question	Landscape
SRQ1	What transformer architectures and techniques have been used for simplifying or explaining clinical text to patients?	Methods
SRQ2	What datasets exist for clinical text simplification and explanation, and what are their characteristics (size, language, construction)?	Resources
SRQ3	How is output quality evaluated, including automatic metrics, human evaluation, and factuality checks?	Evaluation
SRQ4	What approaches extend clinical simplification or generation to multilingual and low-resource settings?	Bridge to Bangla
SRQ5	What Bangla NLP resources and studies exist in the clinical and healthcare domain?	Bangla inventory
SRQ6	Based on identified gaps, what would a Bangla patient-friendly explanation system need to be feasible and factually safe?	Future-work justification

Table 3.3 - Six research sub-questions mapped to their landscape role.

4. PROSPERO Pre-Registration Plan

PROSPERO is an international, free, open-access database of systematic review protocols maintained by the National Institute for Health Research (NIHR) Centre for Reviews and Dissemination at the University of York. Pre-registration is mandatory in this plan because it (i) publicly time-stamps the review protocol, (ii) prevents selective outcome reporting, (iii) reduces duplication by allowing other teams to discover ongoing reviews, and (iv) signals rigor to Q1 journal editors. The registration will be submitted during Week 1, before any database searches begin.

Planned Registration Fields

Field	Planned Value
Title	Factuality in Medical Text Simplification: A Systematic Review with a Focus on Low-Resource and Multilingual (Bangla) Settings.

Field	Planned Value
Review question	See SRQ1-SRQ6 in Section 3.
Search strategy	Five-database Boolean + MeSH search (see Section 5); grey literature via Google Scholar (snowballing only).
Condition / domain	Clinical NLP, patient-facing health communication, low-resource language adaptation.
Eligibility criteria	See Section 6 (Inclusion / Exclusion).
Population	Clinical and biomedical text intended for patient comprehension.
Intervention	Transformer-based simplification, summarization, or explanation generation.
Comparator	Non-transformer baselines, multilingual vs. monolingual, expert gold standard.
Outcomes	Factuality preservation, readability, patient comprehension, cultural appropriateness.
Data extraction	12 fields per study, dual-independent with third-reviewer conflict resolution (see Section 9).
Risk of bias	Cochrane RoB 2 for RCT-style NLP studies, ROBINS-I for non-randomized, CASP for qualitative (see Section 8).
Synthesis	Narrative synthesis under SWiM guidelines; GRADE certainty rating per outcome (see Section 10).
Anticipated registration ID	CRD42026XXXXXX (to be assigned by PROSPERO upon acceptance).
Conflict of interest	None declared. Authors have no commercial interest in any simplification system.
Amendment policy	Any deviation from the registered protocol will be documented as a formal amendment with rationale and date, and disclosed in the final paper.

Table 4.1 - Planned PROSPERO registration fields.

5. Search Strategy (PRISMA-S)

The search strategy follows the PRISMA-S reporting guideline for electronic search strategies, which is the standard for transparent and reproducible literature identification. Five primary databases are searched: ACL Anthology (the primary NLP methods venue), PubMed and PubMed Central (clinical-NLP framing), IEEE Xplore (engineering and digital health), ACM Digital Library (computing surveys), and Scopus (cross-disciplinary snowballing). Google Scholar is used only for grey-literature discovery and snowballing, and is documented as such per PRISMA reviewer norms to avoid the well-known reproducibility issues with its search API.

Core Boolean + MeSH String

Search String S1 - Clinical Simplification (English)
("clinical text" OR "pathology report" OR "discharge summary" OR "medical report" OR "clinical note*" OR "electronic health record*") AND ("simplif*" OR "lay summa*" OR "patient-friendly" OR "plain language" OR "explanation generation" OR "paraphras*") AND ("transformer*" OR "BERT" OR "GPT" OR "T5" OR "BART" OR "mT5" OR "large language model*" OR "LLM*" OR "Mistral" OR "Llama") AND ("factuality" OR "faithfulness" OR "medical accuracy" OR "hallucination" OR "entailment" OR "natural language inference" OR "constrained decoding")

Bangla / Multilingual String

Search String S2 - Bangla and Low-Resource (English + Bangla)

(Bangla OR Bengali OR low-resource language OR multilingual
OR "cross-lingual" OR "South Asian")
AND ("clinical" OR "medical" OR "biomedical" OR "healthcare")
AND ("NLP" OR "OCR" OR "translation" OR "summarization"
OR "named entity recognition" OR "text simplification" OR "simplif*")

Database-Specific Field Tags

Database	Field Tags Used	Notes
PubMed	[tiab], [mh], [pt]	MeSH terms exploded; filters: 2018-2026, English or Bangla.
ACL Anthology	Full text + metadata	Use Anthology API for reproducibility; archive manual queries as JSON.
IEEE Xplore	Command search; metadata fields	Filter: Conferences + Journals; 2018-2026.
ACM Digital Library	Advanced search; Abstract + Title	Filter: ACM SIG conferences prioritized.
Scopus	TITLE-ABS-KEY	Limit to subject area COMP, MEDI, SOCI; document type Article, Conference Paper.
Google Scholar	Full-text query	Grey literature + snowballing only; first 200 results screened.

Table 5.1 - Database-specific search syntax and filters.

PRISMA-S Reporting Checklist Items

The full PRISMA-S checklist (16 items) will be completed and submitted as a supplementary file with the review paper. Key items include: database name and platform, date of search, complete search strategy per database, limits and filters, supplementary search methods (hand-searching, citation chasing), peer review of the search strategy by a second information specialist, and reproducibility notes. The search strategy will be archived in a public Zenodo repository with a DOI.

6. Inclusion and Exclusion Criteria

Criterion	Include	Exclude
Publication year	2018 - 2026	Pre-2018 (foundational works cited separately)
Peer review status	Peer-reviewed OR vetted preprint (arXiv, ACL RollCall)	Non-archival workshop notes without methodology
Language	English or Bangla	Other languages (handled via translation only if pivotal)
NLP component	Must include a computational NLP method	Pure clinical or pedagogy studies without NLP
Domain	Clinical, biomedical, or healthcare	General-domain text simplification

Criterion	Include	Exclude
Task	Simplification, lay summary, explanation generation, medical NLI, NER for healthcare	Machine translation without simplification focus
Bangla relevance	Direct OR transferable	Neither (excluded at full-text)
Full text access	Available via institutional or open access	Inaccessible after 2 requests
Document type	Journal article, conference paper, systematic review	Editorial, abstract only, poster

Table 6.1 - Inclusion and exclusion criteria.

7. Screening Workflow and PRISMA 2020 Flow

Screening follows the four-phase PRISMA 2020 flow: identification, screening, eligibility, and inclusion. After the database searches are executed, results are deduplicated using Rayyan (an AI-assisted systematic review platform) and Zotero. Title and abstract screening is performed independently by two reviewers; conflicts are resolved by a third reviewer. Full-text review is similarly dual-independent. Cohen kappa is computed at each phase and reported in the paper; a target of at least 0.6 (substantial agreement) is required before proceeding to the next phase. Snowballing (forward and backward citation chasing) is performed only after the database search is locked, and the snowballed records are reported separately in the flow diagram to maintain transparency.

PRISMA 2020 Flow - Anticipated Counts

Phase	Anticipated Count	Notes
Records identified (databases)	850 - 1200	Across 5 databases, before deduplication.
Records identified (other)	50 - 100	Grey literature, snowballing, hand-searches.
Duplicates removed	180 - 280	~20-25% duplication rate typical.
Records screened (title + abstract)	700 - 1000	Dual-independent in Rayyan.
Records excluded at T/A	500 - 750	~70% exclusion typical.
Full-text reviewed	150 - 250	Dual-independent; conflict resolved by reviewer 3.
Full-text excluded	90 - 160	With reasons (see exclusion categories).
Studies included in synthesis	60 - 90	Final included set for data extraction.
Cohen kappa (target)	≥ 0.60	Computed at T/A and full-text phases.

Table 7.1 - Anticipated PRISMA 2020 flow counts (estimates based on similar reviews).

8. Quality and Risk-of-Bias Assessment

Risk of bias is assessed using a multi-tool strategy because the included studies span randomized benchmark studies, non-randomized empirical evaluations, and qualitative dataset papers. Cochrane RoB 2 is applied to

studies that report randomized comparisons of simplification systems (rare in NLP but increasingly common with the rise of human preference studies). ROBINS-I is applied to non-randomized intervention studies that compare two or more systems on a shared benchmark. CASP checklists are applied to qualitative papers that report only dataset construction or expert annotation studies. AMSTAR-2 is applied only if any prior systematic reviews are included (for an umbrella review layer).

Tool	Study Type	Domains Assessed	Reporting
Cochrane RoB 2	RCT-style empirical NLP with randomized human preference	Randomization, deviations, missing outcome data, measurement, selection of reported result	Per-study traffic light + summary bar chart
ROBINS-I	Non-randomized comparative NLP benchmark studies	Confounding, selection, classification, deviations, missing data, measurement, reporting	Per-study traffic light
CASP Qualitative	Dataset construction + annotation studies	Aim, methodology, design, recruitment, data collection, rigor, ethics, findings, value	10-item yes/no/can't tell checklist
AMSTAR-2	Prior systematic reviews (if any)	Protocol, selection, data extraction, search, duplication, exclusion, description, funding, conflict	16-item critique
Custom NLP-Bias	Adapted for benchmark-only NLP studies	Train/test leakage, baseline strength, metric validity, statistical testing, reproducibility	5-item score 0-2 each

Table 8.1 - Risk-of-bias tools by study type.

9. Data Extraction Codebook

Data extraction is performed using a structured codebook hosted in a shared Google Sheet with version control. Each included study is extracted by one reviewer and verified by a second; disagreements are resolved by discussion or by a third reviewer. The codebook has twelve fields, each with explicit definitions and controlled vocabularies to maximize inter-rater reliability.

#	Field	Definition and Controlled Vocabulary
1	Citation	APA 7th edition; persistent DOI or ACL Anthology ID.
2	Venue + Year	Journal or conference name; year of publication; quartile if known.
3	Task	simplification summarization explanation paraphrase NLI NER MT (multi-select).
4	Architecture	T5 BART mT5 mBART GPT Llama Qwen Gemma BERT other (free text).
5	Dataset(s)	Name(s), size, language, construction method (expert, synthetic, hybrid).
6	Language(s)	ISO 639-1 codes; flag if Bangla is included.
7	Factuality method	none lexical constraint NER-guided NLI gate graph preservation human-only.
8	Readability metric	FKGL SMOG SARI Bangla-adapted custom none.
9	Baselines compared	List of competitor systems; flag if strong LLM baselines included.
10	Human evaluation	Number of raters, sample size, dimensions, inter-rater reliability (kappa if reported).

#	Field	Definition and Controlled Vocabulary
11	Reported limitations	Authors' stated limitations, verbatim quote allowed.
12	Bangla relevance	direct transferable not applicable; note required.

Table 9.1 - Twelve-field data extraction codebook.

10. Synthesis Methods (SWiM + GRADE)

A formal meta-analysis is not feasible because the included studies use heterogeneous datasets, metrics, and evaluation protocols. Synthesis therefore follows the SWiM (Synthesis Without Meta-analysis) reporting guideline, which structures narrative synthesis transparently across three dimensions: grouping studies for synthesis, describing the standardised metric, and presenting the synthesis. Studies are grouped by (i) factuality-preservation mechanism, (ii) target language family, and (iii) evaluation rigor. Each group is described with a standardised metric (e.g., factuality preservation rate, SARI score where comparable, kappa where reported).

The overall certainty of evidence per outcome is graded using the GRADE approach. Five outcomes are graded: (1) factuality preservation, (2) readability improvement, (3) patient comprehension, (4) cultural appropriateness, and (5) reproducibility. GRADE rates each outcome as High, Moderate, Low, or Very Low based on risk of bias, inconsistency, indirectness, imprecision, and publication bias. A Summary of Findings (SoF) table is included in the final paper.

Thematic Synthesis (Qualitative Layer)

For the qualitative layer (dataset construction, annotation practice, cultural adaptation), thematic synthesis is performed following Thomas and Harden (2008). Line-by-line coding of findings sections is followed by descriptive theme formation and analytical theme generation. Two reviewers code independently; the resulting codebook is reconciled in a structured discussion. NVivo or Taguator is used for coding management.

11. Gap Taxonomy

Five gaps are anticipated based on the scoping search conducted during protocol development. Each gap is formulated to be directly addressable by an empirical study and is mapped to a methodological solution in the companion journal plan.

Gap ID	Description	Evidence in Literature	Solvable by
G1	No expert-validated Bangla clinical simplification corpus exists.	BanglaHealth is paraphrase-only; BanglaCHQ-Summ is summarization; MedSiML-Bn is Gemini-generated synthetic.	Constructing BnClinSimp-Expert (2000 pairs, expert-validated).
G2	Entity Preservation Score (EPS) is conflated with factuality.	Multiple papers report EPS as factuality metric; relation inversions are not caught.	Moving to CFGP (graph-level preservation).
G3	No relation-level medical NLI dataset for Bangla.	MedNLI is English-only; XNLI has no medical specialization; Bangla MedNLI does not exist.	Creating a 500-pair Bangla MedNLI pilot.

Gap ID	Description	Evidence in Literature	Solvable by
G4	No graph-constrained decoding for low-resource simplification.	Edit-Constrained Decoding (Zetsu 2024) is lexical only; TIGER and MemGraphRAG are inference-time, not decoding-level.	Graph-Constrained Beam Search (Algorithm 4).
G5	Human evaluation is underpowered in prior work.	Most Bangla NLP papers use 2-3 raters, no kappa, no sample size justification.	5-clinician study with Fleiss kappa and power analysis.

Table 11.1 - Five anticipated gaps and their empirical solutions.

12. Inventory of Datasets (Realistically Available)

All datasets listed below are publicly downloadable from Hugging Face, ACL Anthology, Mendeley Data, or institutional repositories as of June 2026. Sizes are approximate uncompressed footprints. The total footprint of the recommended stack (rows marked "included") is approximately 5.3 GB, which fits comfortably on a single 16 GB GPU and is mutually compatible because all are documented in English, Bangla, or both, and all are licensed for research use.

Dataset	Lang	Size	License	Source	Use	Status
MultiMSD	EN + 9	200 MB	CC-BY-4.0	ACL Findings 2025	Complex-simple medical pairs (expert-aligned)	Included
MultiCochrane	EN + 5	150 MB	CC-BY-4.0	EMNLP 2023	Sentence-aligned Cochrane abstract simplifications	Included
Med-EASi	EN	80 MB	MIT	Sun et al. 2023	Expert-annotated easy-to-read medical simplification	Included
Cochrane-auto	EN	120 MB	MIT	TSAR 2024	Automatically aligned biomedical abstracts	Included
README	EN	450 MB	CC-BY-4.0	EMNLP Findings 2024	308K term-definition pairs (medical jargon to lay)	Subset (50K)
MedSiML (Bn subset)	BN	350 MB	CC-BY-4.0	SciProdLLM 2025	Gemini-1.5 generated Bn simplifications	Included (with caveats)
BanglaHealth	BN	1.2 GB	CC-BY-4.0	Data in Brief 2025	200K health-domain paraphrase pairs	Included (pre training)
BanglaCHQ-Summ	BN	180 MB	MIT	BanglaLP 2023	2,350 medical QA summaries	Reference
Bangla-HealthNER	BN	90 MB	MIT	Ahmed et al. 2023	31,783 NER samples, 56.13 F1	Included (NER)

Dataset	Lang	Size	License	Source	Use	Status
Bangla-MedER	BN	250 MB	CC-BY-4.0	PLOS ONE 2026	Multi-type medical entity recognition + taxonomy	Included (alignment)
MIMIC-III discharge	EN	1.4 GB	PhysioNet Credentialed	Johnson et al. 2016	Real discharge summaries (English pivot)	Optional (credentialed)
UMLS-CHV (Bn-translated)	EN+BN	500 MB	UMLS License	NLM + custom Bn translation	Consumer Health Vocabulary, Bn-mapped	Included (RAG index)
mT5-base checkpoint	-	1.2 GB	Apache-2.0	HuggingFace	Pretrained seq2seq backbone	Included (model)
BanglaT5 checkpoint	-	1.0 GB	MIT	BanglaLP 2023	Bangla-pretrained T5	Included (model)

Table 12.1 - Dataset inventory with realistic availability. Total included footprint ~5.3 GB.

13. Inventory of Methods and Baselines

Category	Representative Methods	Role in Review
Sequence-to-sequence models	mT5, mBART, BanglaT5, T5	Fine-tuning baselines; small-parameter factuality-constrained candidates.
Large language models	Llama-3-8B, Qwen2.5-7B, Gemma-2-9B, GPT-4	Zero-shot and few-shot baselines; capability ceiling reference.
Constrained decoding	Edit-Constrained Decoding (Zetsu 2024), Lexically-Constrained (Hokamp & Liu)	Factuality-preserving decoding baselines; lexical-only, no relation preservation.
Retrieval-augmented generation	MedGraphRAG, MemGraphRAG, GraphRAG	Knowledge-grounded generation baselines; reduce hallucination via retrieval.
Graph-based factuality	TIGER, Clinical Fact Graph (CFG), FactGraph	Inference-time fact repair; conceptual foundation for CFGP.
Pipeline methods	IndicTrans2 + English TS, NLLB-200 + BART	Cross-lingual cascade baselines; test translate-then-simplify vs. joint.
Readability control	MedReadCtrl (Tran et al. 2025)	Instruction-tuned readability baselines; English-only, no Bangla adaptation yet.
Evaluation methods	MedNLI, XNLI, BERTScore, SARI, FKGL, Bangla Readability	Factuality and quality metrics; flag Bangla-specific gaps.

Table 13.1 - Inventory of methods and baselines mapped to their role in the review.

14. Expected Contributions of the Review

- **C1.** First systematic mapping of factuality-preservation mechanisms specifically in medical text simplification, distinguishing lexical, relation-level, and graph-level approaches with explicit definitions and representative papers for each.
- **C2.** A taxonomy of CFGP-capable approaches, classifying each included method by whether it preserves entities, relations, attributes, or full graph isomorphism, enabling direct comparison with the empirical contribution planned in the journal paper.
- **C3.** A comprehensive inventory of Bangla medical NLP resources with size, license, construction method, and suitability rating, addressing the persistent fragmentation of low-resource language infrastructure.
- **C4.** An explicit gap-to-empirical-study bridge that maps each of the five identified gaps (Section 11) to a methodological solution in the companion journal paper, making the research arc defensible to editors and reviewers.
- **C5.** A GRADE-rated evidence profile for five outcomes (factuality, readability, comprehension, cultural appropriateness, reproducibility), providing a transparent baseline for future systematic reviews and meta-analyses.

15. Four-Week Execution Timeline

Week	Phase	Activity	Deliverable
W1	Setup	Finalize RQs, build search strings, submit PROSPERO, set up Rayyan and Zotero	PROSPERO ID + locked protocol
W2	Search	Execute database searches, dedup, title/abstract screening (dual)	Raw records + PRISMA identification count
W3	Screen	Full-text review (dual), data extraction, risk-of-bias assessment	Extraction sheet + RoB tables
W4	Synthesize	SWiM synthesis, GRADE rating, write first draft, internal review	Review paper draft v1 (7000 words)

Table 15.1 - Four-week execution timeline for the 1-month review.

16. Risk Register and Mitigations

Risk	Likelihood	Impact	Mitigation
Scope creep (review grows beyond factuality)	High	High	Tight scope statement; weekly self-review against PICO + SPIDER frames.

Risk	Likelihood	Impact	Mitigation
Literature explosion (>1500 records)	High	Medium	Strict inclusion criteria; AI-assisted Rayyan triage; snowballing locked until DB search closed.
Bangla paper scarcity (<20 Bn papers)	High	High	Snowballing from Bn NER + Bn MedER + BanglaHealth bibliographies; contact Bn NLP community.
Dual-reviewer availability	Medium	High	Recruit 2 master's students as co-reviewers; offer co-authorship; train on pilot set.
PROSPERO delay (>2 weeks)	Low	Medium	Submit Day 1; use anticipated ID in protocol; begin searches only after acceptance.
Low GRADE certainty (all outcomes Very Low)	Medium	Medium	Transparent reporting; emphasize gap-bridging contribution; propose future RCT-style work.
Venue rejection (JBI)	Medium	High	Backup venues: ACM Computing Surveys, Artificial Intelligence in Medicine; format-flexible draft.
Ambiguous factuality definitions across papers	High	Medium	Working definition (Section 1) applied uniformly; disagreement logged in extraction sheet.

Table 16.1 - Risk register with mitigations.

17. Target Venue and Submission Package

The primary target venue is the Journal of Biomedical Informatics (JBI), an Elsevier Q1 journal with a 2024 impact factor of 4.5 and a typical first-decision time of 6-8 weeks. JBI scope explicitly includes clinical NLP, health communication, and systematic reviews, making it a strong fit. The journal allows systematic reviews up to 12,000 words and requires PRISMA 2020 compliance. A backup venue is ACM Computing Surveys (Q1, IF 16.6) which accepts systematic reviews but with a broader computer-science scope; a second backup is Artificial Intelligence in Medicine (Q1, IF 7.0).

Submission Checklist

- Main manuscript (7000-9000 words excluding references and appendix).
- PRISMA 2020 main checklist (27 items) as a supplementary file.
- PRISMA-S search reporting checklist (16 items) as a supplementary file.
- Search strategy appendix (one full Boolean + MeSH string per database).
- Data extraction sheet (CSV or XLSX) with all 12 fields per included study.
- Risk-of-bias traffic-light plots (RoB 2, ROBINS-I, CASP) as supplementary figures.
- GRADE Summary of Findings (SoF) table for the 5 outcomes.
- PROSPERO registration ID cited in the Methods section.
- Cover letter highlighting the 5 contributions (C1-C5) and the gap-to-journal bridge.
- Conflict of interest and funding statement (none declared).

18. Expected Review Paper Outline

Section	Words	Content
Abstract	250	Background, methods (PRISMA, PICO+SPIDER, GRADE), key findings, conclusions.
1. Introduction	800	Motivation, problem statement, contribution summary, paper structure.
2. Methods	1200	Protocol, PROSPERO, search strategy, screening, RoB, synthesis (SWiM+GRADE).
3. Results - Flow	400	PRISMA flow diagram + counts, kappa values.
4. Results - Methods Landscape	700	SRQ1 answer: architectures and factuality mechanisms.
5. Results - Resources Landscape	600	SRQ2 + SRQ5 answer: datasets + Bangla NLP inventory.
6. Results - Evaluation Landscape	500	SRQ3 answer: factuality metrics, human eval rigor.
7. Results - Bridge to Bangla	400	SRQ4 answer: multilingual / low-resource approaches.
8. Discussion	1500	Taxonomy, gap analysis (G1-G5), bridge to empirical paper, limitations.
9. Future Work	600	SRQ6 answer: what a Bn system needs; concrete research agenda.
10. Conclusion	250	Restate contributions; call to action.
References	n/a	80-120 references; APA 7th edition.
Appendix A	n/a	Full search strings per database.
Appendix B	n/a	Data extraction sheet (sample).
Appendix C	n/a	GRADE SoF table.

Table 18.1 - Planned review paper outline with target word counts.

19. References (Anchor Papers)

Fifteen anchor references that frame the review methodology and the empirical gap analysis. Full bibliography (80-120 references) will be assembled during the search.

- [1] Page, M. J., McKenzie, J. E., Bossuyt, P. M., et al. (2021). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*, 372, n71.
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